TARGETED PROFANITY LABELER REPORT

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Introduction / Motivation

Targeted profanity is a persistent issue on all social media platforms, including Bluesky, showing up across a diverse range of communities and user groups. While general profanity can be harmless in casual contexts (e.g., “oh sh*t, i forgot to do my homework”), its directed use (e.g., profanity aimed at specific individuals/groups of people) creates a hostile environment. Many users may find this kind of language offensive or emotionally draining, and it plays a role in pushing users out of conversations they might otherwise engage in. It’s also of concern for parents who allow their children on social platforms but might want a tool that can filter out explicit language entirely. Beyond that, many adults on Bluesky simply want to avoid toxic interactions and prefer to hide messages that use profanity as a way to demean or belittle others. By addressing targeted profanity specifically, we hope to reduce harm by supporting healthier conversations on Bluesky and giving users the tools to control what they see on the platform.

Approach / Policy

Our approach focuses on giving users flexible control over the level of profanity they want to encounter. The labeler classifies each post into three categories (targeted profanity, general profanity, or neutral) and assigns clear labels to the first two: “💩 talker” for posts using targeted profanity directed at a proper noun (e.g., a curse word + @ mention + negative sentiment) and “🤬” for general profanity pulled from a list of known curse words in the English language ([bad-words.txt]). Clean posts are labeled internally and do not display any icon. Users can then choose whether posts in each category are hidden, blurred with a warning, or shown normally. Given more time, we would hope to add action at the account level where users who frequently engage in profane speech can be filtered with “block,” “warn,” or “show” options, and their profiles will display a “potty mouth” banner to signal repeated misuse. We hope that this will discourage users from repeatedly creating harmful posts while still preserving freedom of expression. By making these behaviors visible and actionable at both the post and account levels, we can create social accountability without imposing strict platform-wide restrictions that not all users may want. Ultimately, this structure balances user safety, transparency, and customizable control over content exposure.

Data Collection and Annotation

We began collecting training data by finding a list of known profane words (and their misspellings) and used those as our keywords. These words and the list were obtained from the bad-words.txt provided by CMU. We then retrieved approximately 50 posts per keyword from Bluesky using the platform's API. The collection process retrieved not only post content but also contextual information, such as whether or not it was a quoted, mentioned, or reply post. From this dataset, we randomly selected 150 posts for human ground-truth annotation and an additional 5,000 posts for model training (annotated using Llama3.1), while ensuring that there was strict separation between these subsets to prevent data leakage. We also ensured that the 150-sample test set contained a roughly balanced distribution of the three classes (targeted, general, neutral), and included posts with mentions, quotes, and replies to avoid structural bias. At first, the ground truth annotation process revealed that there were disagreements among team members regarding label decision criteria. However, this revelation was essential since we were able to expose ourselves to individual edge cases. For instance, teammates disagreed on whether slandering a movie using profanity counted as targeted profanity or not. After a more intensive discussion, we developed more rigorous labeling definitions, and three of our team members annotated an additional randomly selected 150 posts. We resolved disagreements through real-time discussion and majority voting. Due to time constraints, we were unable to expand beyond this 150-sample annotated dataset.

Two-Layer Labeler Design

Our labelling system uses a two-layer classification approach: Layer 1 determines whether or not profanity is present in a given post. Layer 2 determines whether the identified profanity constitutes a targeted attack. For profanity detection, we later implemented a multi-model ensemble after determining that simple sentiment analysis produced excessive false positives by misclassifying expressions of sadness or depression. Our final version of Layer 1 combines Toxic-BERT for efficient detection of offensive language, NLLB-small100 for translating non-English content, and BERT Large Adult Text Classifier to capture sexually explicit content that Toxic-BERT tends to miss.

For Layer 2, we addressed the semantic complexity of targeted attack detection through knowledge distillation. We first considered using commercial LLM APIs such as Gemini and OpenAI. However, these services incur token costs and frequently refuse to respond due to the sensitive nature of the content. We also tested Llama3.1:7B locally via Ollama, which functioned correctly, but its large size (42 GB) led to extremely slow inference and made it impractical for TA testing or real-world deployment. To resolve this, we fine-tuned microsoft/deberta-v3-base and performed knowledge distillation, using Llama3.1 as the teacher model. DeBERTa-v3, developed by Microsoft Research, uses a "Disentangled Attention Mechanism" to model content and positional information separately, allowing it to better capture complex semantic structures than traditional BERT-based architectures. In our task, the model must differentiate between *the original post* and the *quoted/replied content*, making DeBERTa particularly well-suited to the problem. The distillation process involved generating labels for 5,000 training samples using Llama3.1 over approximately 10 hours, then fine-tuning DeBERTa-v3 on these outputs. As part of the distillation process, the DeBERTa-based student model was able to achieve over 89% precision and 90% recall in replicating the teacher model's behavior. This suggests that our selected contextual features—such as author identity, quoted text, mentions, and reply chains—are strongly correlated with targeted attack behavior.

Our system was fully modularized into an end-to-end pipeline, consisting of separate modules for feature extraction, context construction, multilingual translation, multi-model inference, and class merging. Each component can be independently tested and replaced, enabling scalability and future improvements.

Test Results

On the 150 manually labeled test samples, our system achieved an overall labeling accuracy of approximately 72%. The profanity classification layer demonstrated strong performance with 90% precision and 88% recall, while the targeted attack classification layer achieved 81% precision and 85% recall. Computational efficiency proved suitable for real-world deployment, with the complete labeling pipeline processing the test set in 104 seconds of wall-clock time (33.52 seconds user CPU time, 12.49 seconds system CPU time, 44% average CPU utilization), translating to approximately 0.7 seconds per post. Given the subjective nature of profanity and targeted attack detection (tasks that rely heavily on nuanced semantic interpretation and contextual understanding), we believe these results represent a reasonable balance between effectiveness and deployability under our time and resource constraints. The moderate accuracy loss is acceptable for deployment on Bluesky since it can be mitigated through additional human review and user feedback, which will allow the system to improve over time.

Analysis of Results

These results reflect the subjective nature of the task, where even team members initially disagreed on labels. Detecting general profanity is straightforward, while identifying targeted attacks requires understanding intent and context, which explains the higher precision in the first layer. For our use case, high precision is critical to avoid overwhelming users with false positives and maintain trust. Recall is also important for safety, but a slightly lower recall is acceptable because users and moderators can review borderline cases, enabling system improvement over time. The current balance, strong precision with slightly lower recall, fits our goals well, allowing the system to act as a first-line filter while minimizing unnecessary flags. The labeling pipeline is computationally efficient, processing posts quickly enough for real-world deployment and scaling without excessive resource use.

For future improvements and deployment implications, we encountered several challenges that need to be addressed before large-scale deployment. The system struggles with non-English content, incorrectly flags proper nouns like place names (e.g., "Punta Cana") or usernames, and doesn't understand cultural differences in profanity (e.g., British people using "bloody" casually). We also found that Llama3.1 was too sensitive and created too many false positives, which we had to manually adjust. Future work should add severity levels to separate mild words like "damn" from serious slurs, expand the annotated dataset beyond 150 samples to cover more edge cases, and build feedback systems where user reports improve the model over time. From an ethical perspective, we need to consider potential misuse; some users might try to earn the "potty mouth" badge on purpose, and whether automated systems can properly handle linguistic and cultural diversity. We believe this labeler works best as a tool that gives users control over their own feed rather than enforcing strict platform-wide rules. This approach balances user safety with freedom of expression on Bluesky.